

Executive Summary

Project Title

Fixing the Gaps: A Stakeholder-Driven Power BI Dashboard for Resource Allocation and School Performance in Pakistan

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This project develops a stakeholder-driven Power BI dashboard to support evidence-based education planning and resource allocation in Pakistan. The analysis combines school infrastructure, enrollment, education performance, retention, learning outcomes, geographic information, and socio-economic/security context to identify regions where intervention is most needed.

The dashboard was designed for government education decision-makers and provides multiple levels of analysis. The **Executive Governance** view provides a high-level summary of education performance and resource conditions. The **Executive Risk & Intervention** view identifies regions with critical gender and infrastructure risks. The **District Infrastructure Audit/Analysis** view investigates infrastructure deficits at province and district level. The **Policy Impact & Socio-Economic** views examine relationships between educational investment, infrastructure, socio-economic context, terrorism indicators, and education outcomes.

The analysis identifies substantial differences between regions. ICT records an average education score of **81.3**, learning score of **64.6**, and retention score of **81.3**, while Sindh records **55.9, 41.6, and 48.8**, respectively. Balochistan records **51.0, 38.3, and 43.0**, and FATA records **47.5, 37.3, and 38.8**. These differences indicate that education performance is highly uneven across regions.

Infrastructure indicators also reveal significant gaps. Across the overall dataset, average electricity coverage is **47.83**, drinking-water coverage is **59.68**, and toilet coverage is **57.29**, while the overall no-facility rate is **10.26%**. District-level analysis shows that several areas experience much higher deficits. Tharparkar has a **52.89% no-facility rate** and a **33.1 retention score**, while Thatta has a **44.42% no-facility rate** and a **24.3 retention score**.

The risk analysis identifies **Kohistan** as a particularly important intervention region. It has a **93.0% gender enrollment gap**, **37.89% utility deficit**, **28.6% single-teacher strain**, **20.5% single-**

classroom strain, 42.1 learning score, and 30.0 retention score. The combination of these indicators supports a targeted intervention rather than a single-issue response.

The project also identifies important differences in trends. AJK's education score increased from **66.2 in 2013 to 78.1 in 2016**, while Balochistan improved to **57.2 in 2014 before falling to 46.1 in 2016**. This demonstrates why policymakers should examine trends over time instead of relying only on a single year's performance.

The main recommendations are therefore to prioritize integrated interventions in critical regions such as Kohistan, target infrastructure and education support toward high-deficit districts in Sindh and Balochistan, and study successful regions such as AJK and ICT to identify practices that can be adapted elsewhere.

Overall, the dashboard transforms fragmented education data into a decision-support solution that helps stakeholders answer three questions: **Where are the largest gaps? What indicators are associated with weak outcomes? And where should limited government resources be prioritized?**